Prof. Brit Ross Winthereik

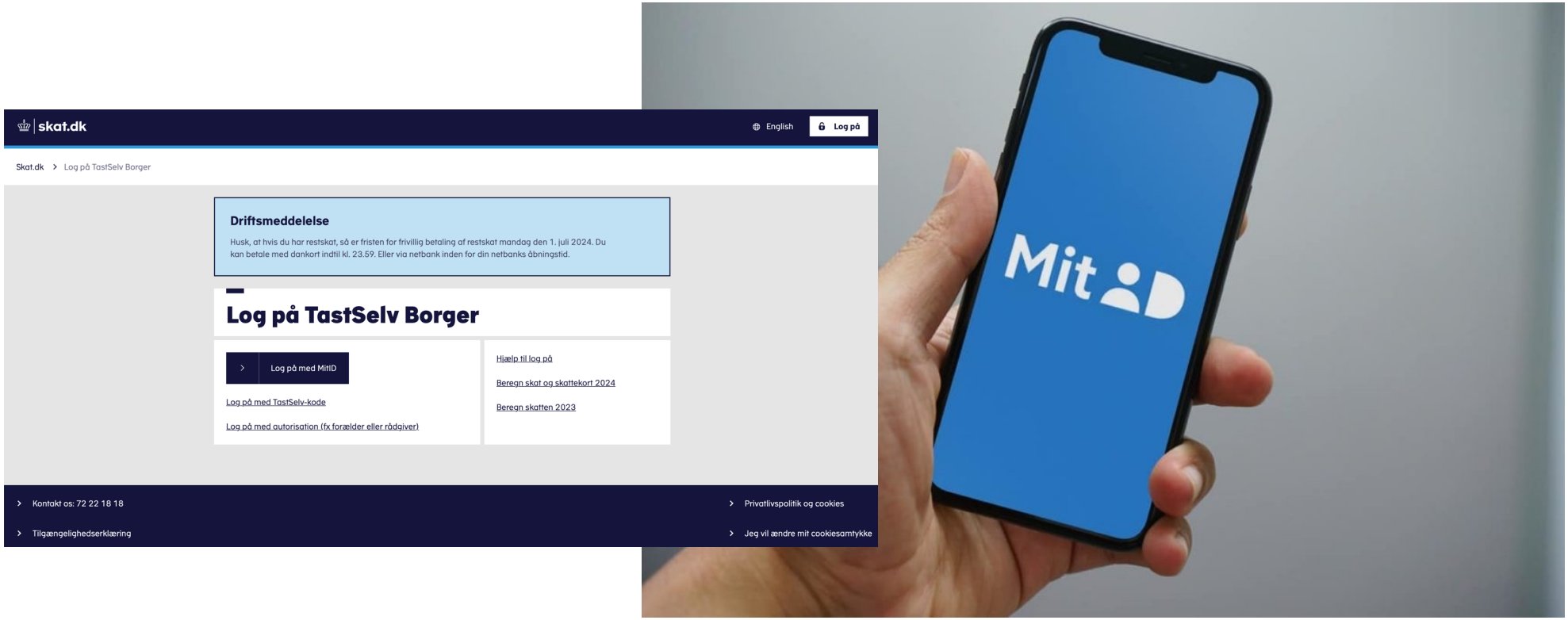
The organizational context

The overall problem

- Many data, AI, and ICT projects *work technically* but fail *organizationally*

- Engineering solutions often assume:
  - Better data → better decisions
  - Better systems → better behavior

- **Claim**: Technical systems and analytics do not create better performance on their own – organizations do!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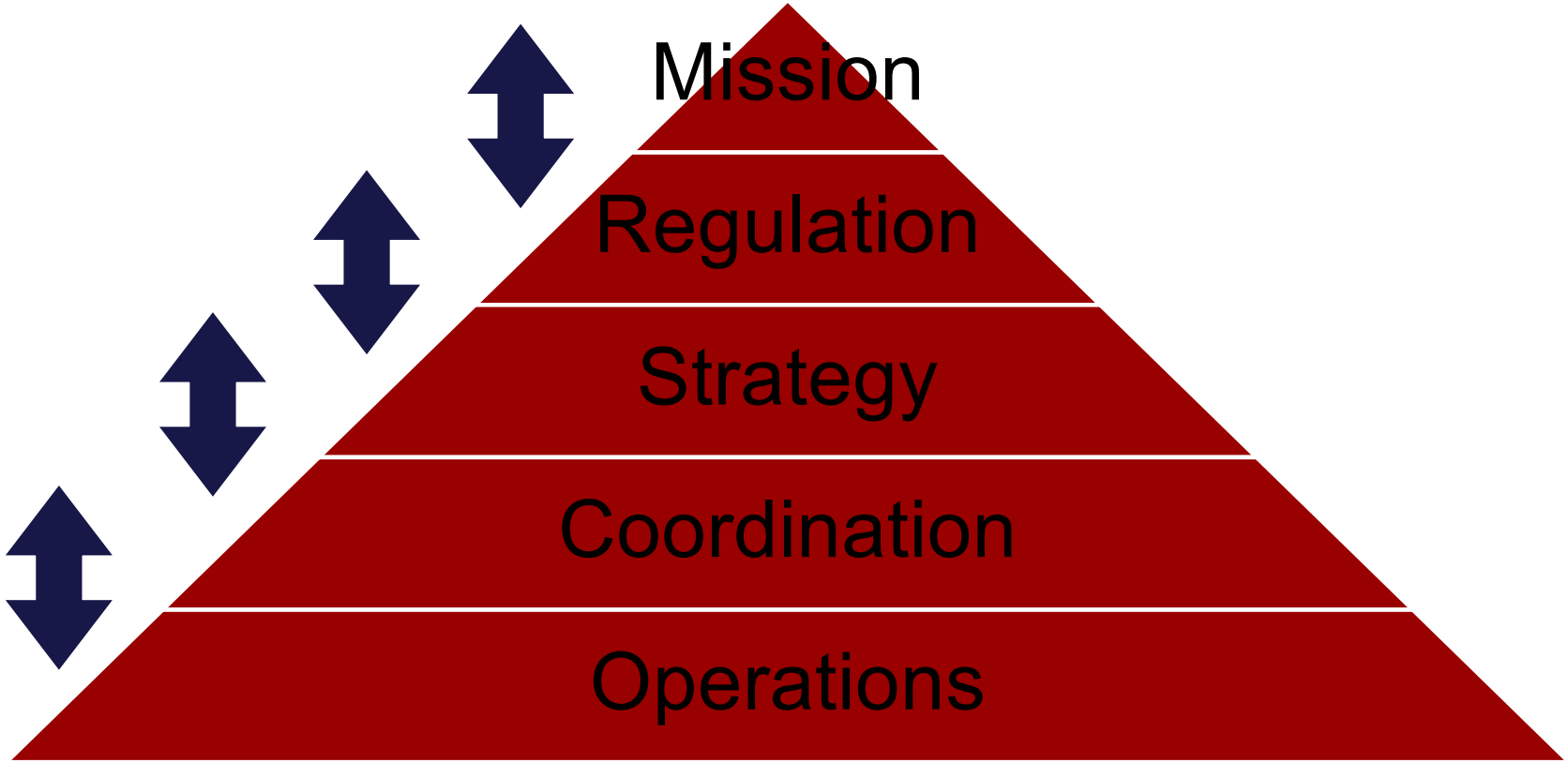
The organisational challenge



The accountability sink

Remember the organization when fixing immediate problems in the data-driven organization

Phishing emails account for 15% of all data breaches

Problem

People stop clicking and reduce the risk of hacker attacks

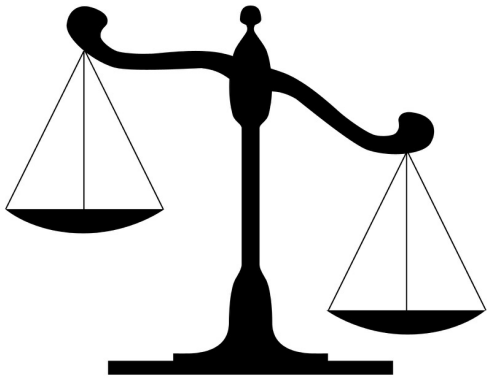
Solution

Possible actions and trade offs

- Put on hefty spam filters **at the risk** that employees loose out on information.
- Punish people who do it **at the risk** of loosing trust in management.
- Train them to recognize it **at the risk** of efficiency.
- Identify the segment who has the riskiest behavior **at the risk** of wrongfully shaming individuals.

Definition of “trade-off”

“A **decision** where you give up one thing in order to gain something else, often involving a balance between different factors. It reflects the idea that choosing one option typically means sacrificing another.”



Today's lecture

1. Start with **three real organizational examples** to illustrate the role of organizational context for performance and technology use.

2. Introduce how to use core concepts: **affordance, invisible work, organizational fabric** to understand the organizational context.

3. Explain how to move from **understanding the organizational context to better decision-making.**

Organization and technology co-produce each other

Co-production refers to the simultaneous processes through which **society and scientific knowledge** influence and shape each other.

In a company it emphasizes that **everyday norms and routines** and **systems** evolve together, highlighting the interconnectedness of organisation and technology.

(Jasanoff, 2004, States of Knowledge: The Co-Production of Science and Social Order)

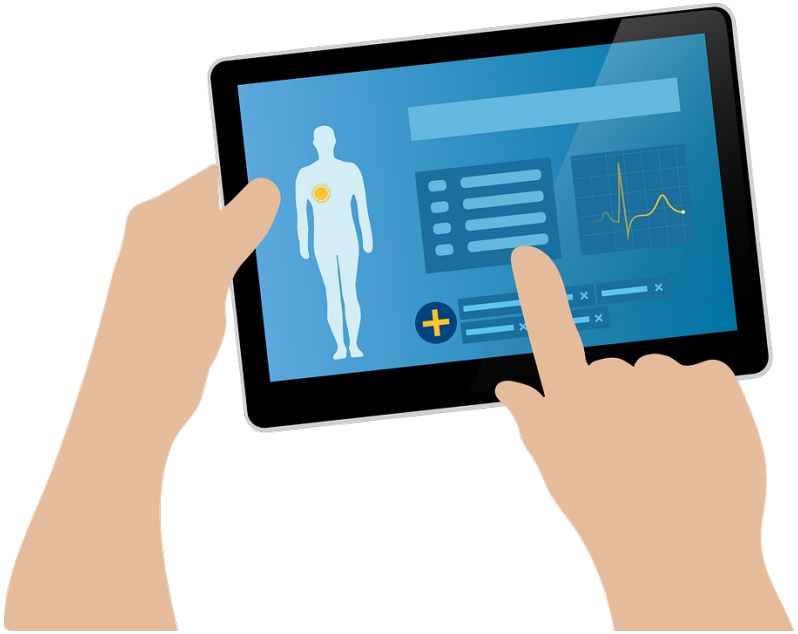
Example 1: Affordance

- **Definition** of affordance: Is what the environment **offers** to a person or a group of employees.

- **Case:** To use data managers must trust data, but data are entered by different professional groups, who use different standards and have different routines.

- How to turn the electronic medical record into an environment for **trustworthy data?**

- Consider the **relation** between technology, users and organizational context, ie standards and work routines.



Exercise: Affording trust in data

- Trust in data is crucial for the hospital management to use data for decision-making but the data are entered by different professional groups using different standards and routines. As a result, the data are not trusted.
- Think of one change to either standards or work routines to afford trust in data.

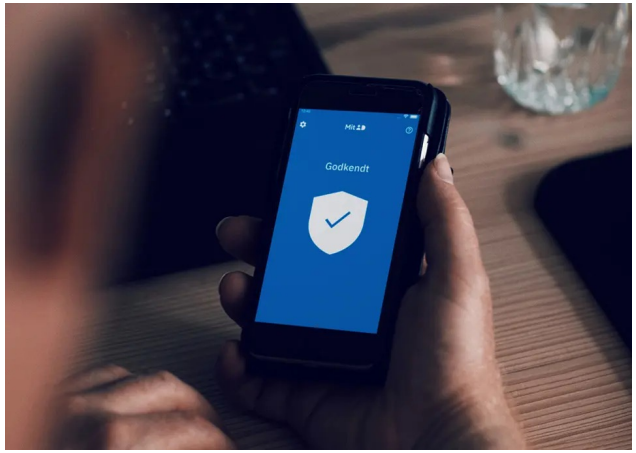
Example 2: Invisible work

- **<u>Definition</u>** of invisible work: Activities that are informal, unaccounted for and under-valued, but nevertheless keeps organizational performance on track.

- **<u>Case:</u>** Connecting data sets, data cleaning, compensating for missing data to keep systems running and enable data quality.

- How to support these invisible tasks? If work is invisible, it cannot be designed for?

- **"Accountability drains**" and lack of organizational learning and innovation.



Exercise: Supporting invisible work in data analysis

- Consider the invisible work involved in preparing data for analysis.
- How might managers better recognize and support such tasks without knowing the exact details (of this work, don't propose making it visible)?
- How can organizational elements such as time allocation, coordination, and collaboration play a role?

Example 3: The organizational fabric

- **Definition** of organizational fabric: Intertwined social and technical threads to understand organizations through organizing processes.

- **Case:** Algorithmic management systems in large companies spanning multiple sites inefficient.

- Implementation of technology is an **ongoing process** that makes organizations.

- Consider the impact of **power structures** (different positions in hierarchies) **and access to technologies** on organizing.

Example 3: The organizational fabric

- Consider the dashboard as a tool for managers to gain overview and monitor company performance.
- Using the case example what risks are involved in relying on dashboards for real-time monitoring of tasks?
- What can be done to mitigate those risks?

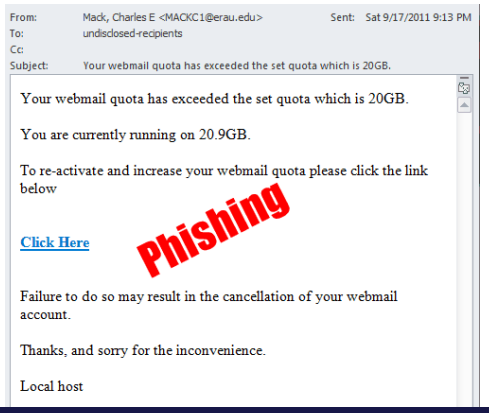
Technological fixes won’t save bad company performance

- When technology does not work as a solution to a problem, the problem may be socio-technical.
- “a technical fix” – often used to explain attempts to solve a sociotechnical problem, where there is little attention to the organizational or social elements of the problem.
- A socio-technical approach attends to
  - Processes
  - Relations across technology, people, standards etc.

Redesign relation between problem and solution

Problem

- Little trust in data
- Lack of innovation
- No productivity gains



Solution

- Training of employees
- Change expectations
- Allow for variation
- Introduce quality measures across the organization
- Feedback mechanisms to ensure organizational learning
- Different workflows
- Attention to power structures